ICE / POWERTRAIN · 0.8.5 · ENGLISH CALCULATION DOCUMENT

Powertrain vehicle

Combustion engine — comparison configuration

From component calibration to traction demand, power flow and battery state-of-charge change.

10,81 kW	0	0 pp
Wheel demand at the calculation point	kWh/100 km, net at terminals	SOC change over the prescribed mission

> The mission is numerically supplied, but the powertrain model lacks complete experimental validation. **Outside digitised domain / generating assumptions: 0 of 600 s.** Installed component packages: 8. Their evidence status is shown separately.

The 720° engine calculation remains separate: ice.html. Exporting this document includes the current parameters and results, rather than an outdated baseline PDF.

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0. Engine inputs and optional vehicle selection

The calculation starts with prefilled, editable fields. The initial example is the 1ZR-FAE combustion engine with a bounded 97 kW / 160 N·m envelope. Geometry, pressure and forces remain in the separate 720° module. Selecting a vehicle is optional.

Engine, electric-drive and vehicle parameters are kept separate. Combustion-engine power, torque and speed fields constrain the source envelope; they do not create a new measured characteristic. Raising a limit beyond the source boundary adds no power. A different engine requires its own data.

Selecting a profile or using the advisor first displays the proposed changes. Manual editing protects a field from replacement; that protection can be cleared. Changes are applied only with a separate button. The most recent application can be undone.

Vehicle profiles are author-defined starting assumptions, not production-vehicle specifications. The advisor does not relax requirements to declare a configuration suitable. A missing 1ZR fuel map means unknown consumption, not zero.

Initial-configuration parameter	Value
Layout	Combustion engine — comparison configuration
Combustion engine / source	1ZR-FAE — two-point output-torque alignment
Entered power / torque limit	97 kW / 160 N·m
Entered speed range	1500–6400 min^{-1}
Mass / wheel radius	1600 kg / 0.31 m
Fuel-consumption status	No map — value unknown
Replacement	Profile → preview → apply unprotected fields

1. The powertrain as a system

The top calculation level is the vehicle and its powertrain, rather than an isolated combustion engine. Supported layouts are battery-electric drive (BEV), a P2 parallel hybrid, a series hybrid and a conventional combustion engine for comparison. The existing 720° combustion-engine calculation remains separately available in ice.html; it has not been replaced by an efficiency table.

In P2, the combustion engine and electric machine act on a common shaft upstream of the gearbox. At a given vehicle speed, their torques can be added only when rotational speeds, transmission ratio and battery limits are compatible. Rated maximum powers cannot be added independently of speed. In a series layout, only the traction electric machine is mechanically connected to the wheels; the engine supplies a generator and the DC bus [9].

This is a quasi-steady backward calculation: prescribed v(t), gradient and vehicle parameters determine the required force. Powertrain limits are checked at every step. If traction force or auxiliary demand cannot be supplied, the deficit is retained and the mission is marked incomplete. Prescribed speed is not reported as an actually achieved speed.

The configurations are engineering scenarios, not models of a production Outlander or Toyota Hybrid. The baseline mass of 1950 kg, Cd = 0,33 and A = 2,65 m² are example values taken from the library configuration; the new transmission ratios, battery, controls and electric machines do not reproduce that vehicle. Using the same mass deliberately isolates the effect of architecture; the mass of the actual battery pack and assemblies must be accounted for separately [2,3].

No claim is made of Euro 5/6 type approval, test-bench validation of the complete powertrain or a certified driving range. Having a new engine or a catalogue peak efficiency does not establish its full map, controller calibration or thermal capability.

2. Calibration and limits of the evidence

For the 1ZR-FAE, only a two-point alignment of output torque was performed: $M_{кор}(n) = M720(n)\cdot(a + b\cdot n)$. The reference conditions are 160 N·m at 4400 min^{-1} and 97 kW at 6400 min^{-1}, corresponding to 144,732 N·m at the second point. Two coefficients were determined from two points; no independent verification conditions are available. This aligns the output by interpolation; it does not identify combustion processes [1].

Extending the correction without limits to low speeds exceeds the rated maximum torque. Therefore explicit limits of 160 N·m and 97 kW are used for preliminary selection. The portion below 4400 min^{-1} is an assumed bounded envelope; its plateau is not a measured Toyota characteristic. The uncorrected result and the unbounded correction remain in the calibration artifact.

Pressure, temperature, heat balance, fuel consumption and dynamic loads in the 720° module remain results of the original uncalibrated model. The aligned 1ZR envelope has no known fuel map: the application returns “fuel map unavailable” rather than scaling fuel consumption with torque. This preserves the distinction between a power-selection characteristic and a physical cycle.

The Outlander GG2W is the user's own experimental project with an OBD-II logger (author clarification dated 24.09.2026). According to the source README, the Engine Power channel is calculated by the MX+ application, so it is not test-bench power. In the Willans artifact, the input is named engine_power_kw_obd and the target is fuel_g_s_maf: fuel flow derived from MAF/(14,1·λ), rather than an independent fuel-flow meter. This dataset is suitable for a vehicle-specific empirical model, but error relative to this reference is not the absolute error in physical fuel flow [2].

The GG2W Willans coefficients were copied unchanged from vehicle_model_library_examples_v0.11: ṁf = 0,0700027543 + 0,0655540723·Pe, with ṁf in g/s and Pe in kW. At Hu = 43 MJ/kg, the incremental efficiency implied by the slope is approximately 35,48%. The regression intercept of 0,0700 g/s is not a measured idle fuel flow: an assumed 0,22 g/s is specified separately [2]. The application uses an explicit low-load fuel-flow floor of max(0,22; b0+b1Pe) g/s; it prevents consumption from falling below the assumed idle value and was not calibrated with the library coefficients.

In the supplied artifact, integrated fuel error against the MAF/lambda estimate on the test set is −3,088% for the fuel model alone, but −66,565% for the complete hybrid chain. The latter is not acceptable validation of the vehicle model. The original raw trips are absent from the package used; these metrics are reported by the library, not a validation rerun by us. The controller from that chain was not adopted as a verified controller [2].

For retraining on new observations, fit-willans.mjs accepts CSV columns group_id,power_kw,fuel_g_s,dt_s. Whole groups/trips are separated; every fifth ordered group forms the holdout set. Time-weighted squared error is minimized subject to nonnegative coefficients; at least 5 groups, 20 points and variation in power are required. P < 1 kW, nonnumeric values and invalid data are rejected with a row log. A new fit does not replace the previous one automatically.

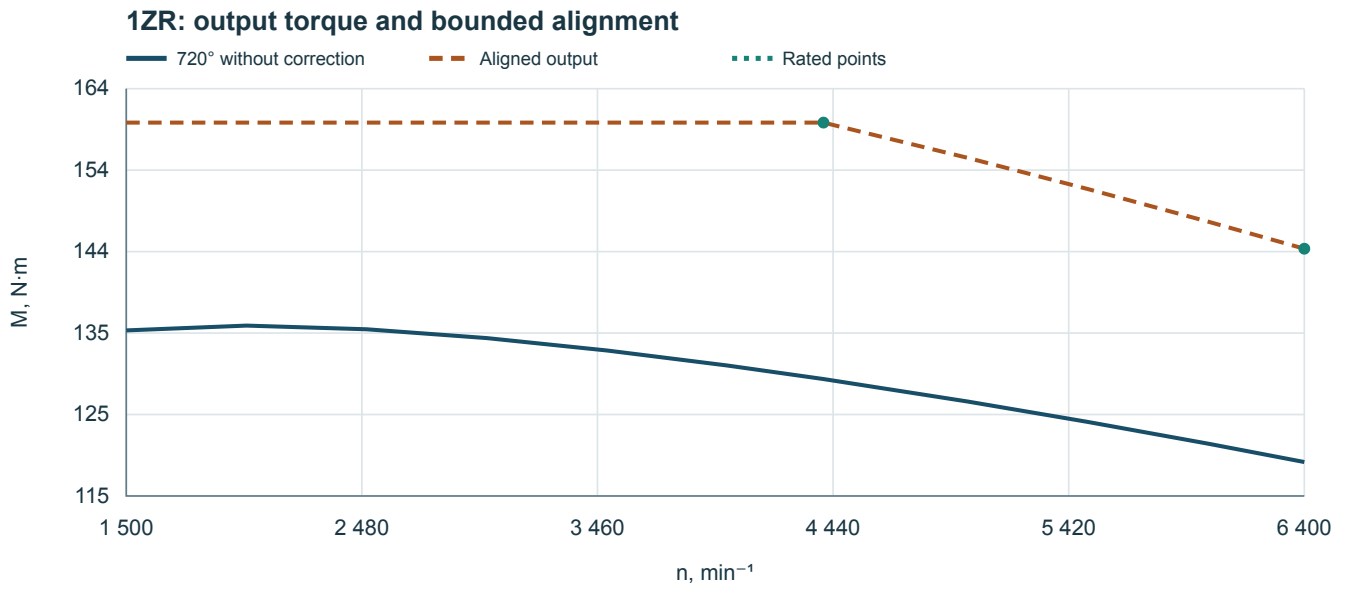
The source diagnostics explicitly state that energy reserve, Battery Charge / Save modes and charging cannot be established without correct traction-battery SOC or a mode flag. Excess calculated engine power may include auxiliary loads, warm-up and channel error. Battery R_{int} and regeneration in this application are separate explicit simplifications, not claimed to be calibrated against these logs [2].

Two-point 1ZR alignment

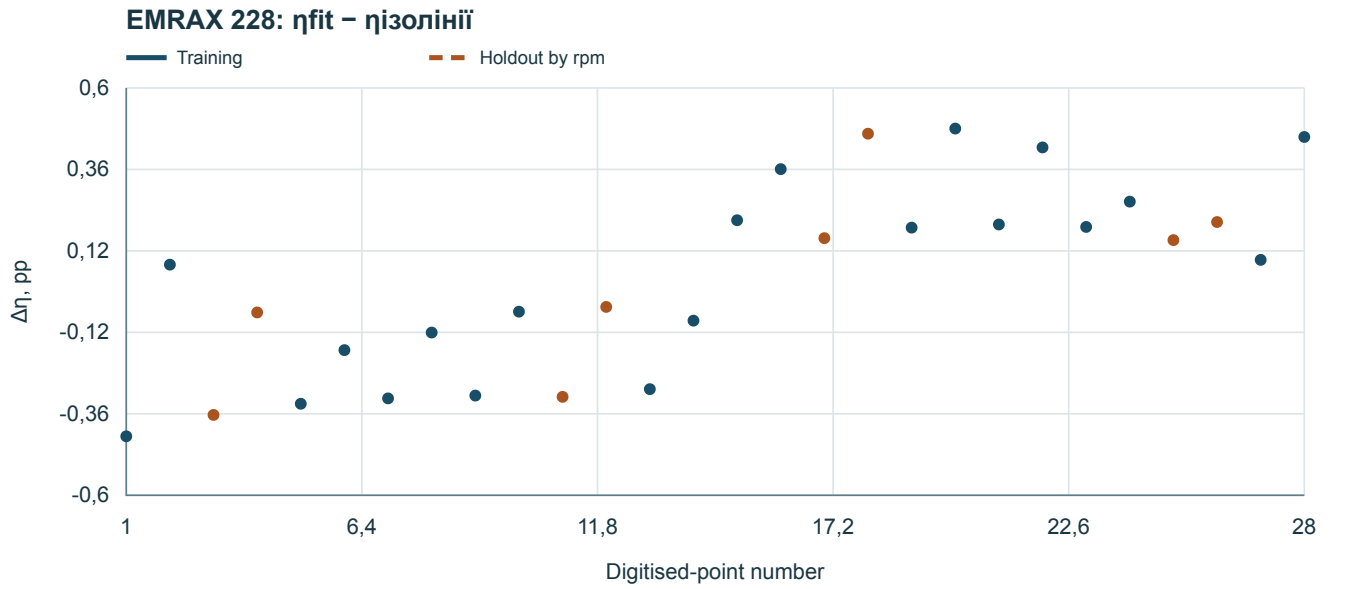
n, min⁻¹	To, N·m	After, N·m	Reference, N·m
4 400	128,951	160	160
6 400	118,904	144,732	144,732

Checks performed, and what no

Source / method	Training	Check	Status
1ZR, two reference points	2 points / 2 coefficients	0 independent points	Output torque only; not fuel/pressure
EMRAX 228, contours	20 points; RMSE 0,284 pp	8 points; RMSE 0,26 pp	Chart fit, not test-bench validation
GG2W fuel, author OBD experiment	Transferred coefficients	−3,088% relative to MAF/lambda	Source metric; not an independent fuel meter
GG2W complete chain, historical report	Controller not transferred	−66,565% relative to the report reference	Not logger accuracy; not a test of the new Powertrain
Brogen contour CSV	Trial fit	See brogen-audit.json	Not applied by default



Two-point alignment of 1ZR output. This transformation does not calibrate pressure, temperature or fuel consumption in the 720° model.



EMRAX 228 contour-fit residuals in percentage points, not relative percent.

3. Combustion-engine and electric-machine maps

EPA describes constructing full ALPHA maps from experimental and modeled regions, including operating-domain definition, separate low-load conditions and verification. A vehicle model needs fuel flow as a function of speed and torque; BSFC and efficiency are derived quantities, not substitutes for idle fuel flow. Cold-start and transient corrections are outside this steady-state implementation [7].

The official EPA page also provides electric-drive maps. EMOT includes the motor and inverter; EDU additionally includes the transmission; BISG includes the belt drive. An inverter or gearbox must not be counted twice. The current importer accepts motor-only and motor+inverter maps. EDU/BISG maps are rejected until a separate adapter for their mechanical boundary is implemented [8].

EPA numerical ZIP packages were not downloaded for this release: methodology pages and the link catalogue were checked, but not numerical matrices. None of the built-in maps is described as an EPA map. The powertrain-grid/1 importer accepts prepared JSON with rpm, torqueNm and values; rows are speeds and columns are torques. This is our exchange format, not a claimed native ALPHA format.

For fuel_g_s, Hu and maxTorqueNm are required at every speed. For loss_w, the system boundary is mandatory; negative torques have their own cells. Zero is a numeric zero; null is missing data. Bilinear interpolation stays inside the map and does not turn missing cells into zeros. At an exact grid node, neighboring cells with zero weight have no effect.

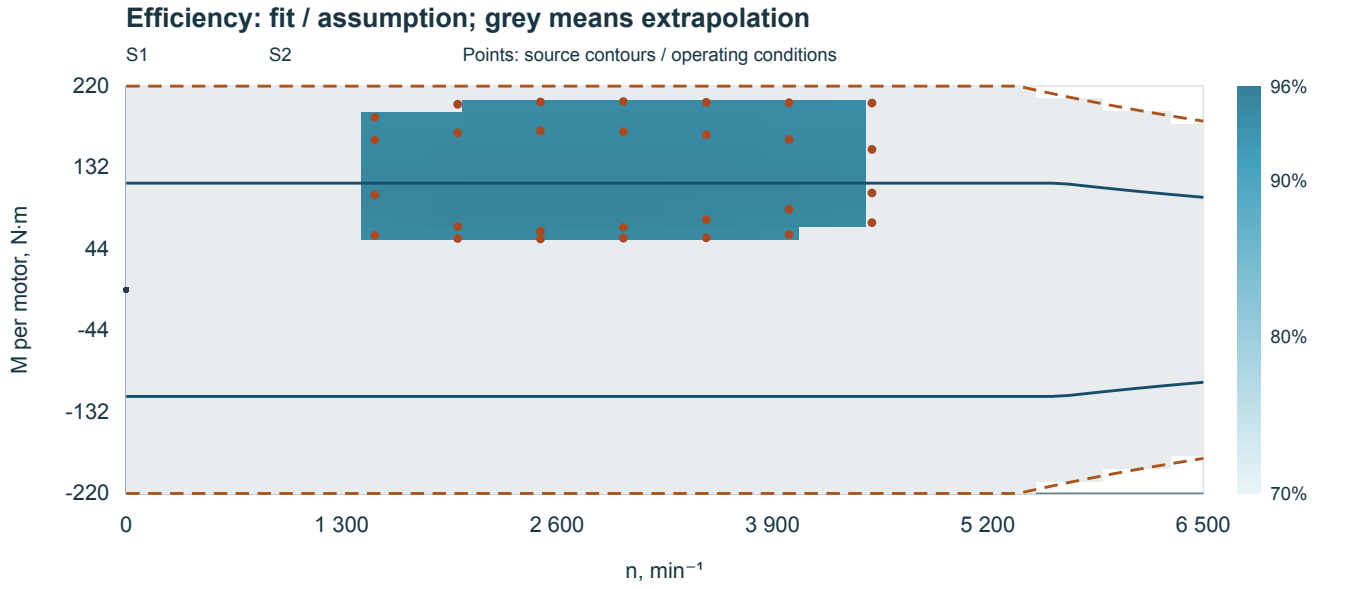
For the EMRAX 228, 28 points were digitized from the 94% and 95% efficiency contours on page 1 of the supplied datasheet. Nonnegative coefficients were fitted in loss space: $L = c0 + c1v + c2v^2 + c3v^3 + c4\tau^2 + c5\tau^2 v$, $v = n/1000$, $\tau = |M|/100$. Training used 20 points; 8 points at 2000 and 4000 min^{-1} were withheld from fitting. Holdout efficiency RMSE is 0,260 percentage point. This checks an approximation of the chart, not a new measurement [4].

A loss floor of $3R25(M/Kt)^2$ and an efficiency ceiling of 96% are used. Estimated digitization uncertainty is ±3 N·m and ±50 min^{-1}; map test conditions are not fully specified. The voltage and cooling of the tabulated MV/LC version do not establish that the chart used identical conditions. Results outside the digitized branches and in generating operation are marked as extrapolation; time spent in these conditions is accumulated.

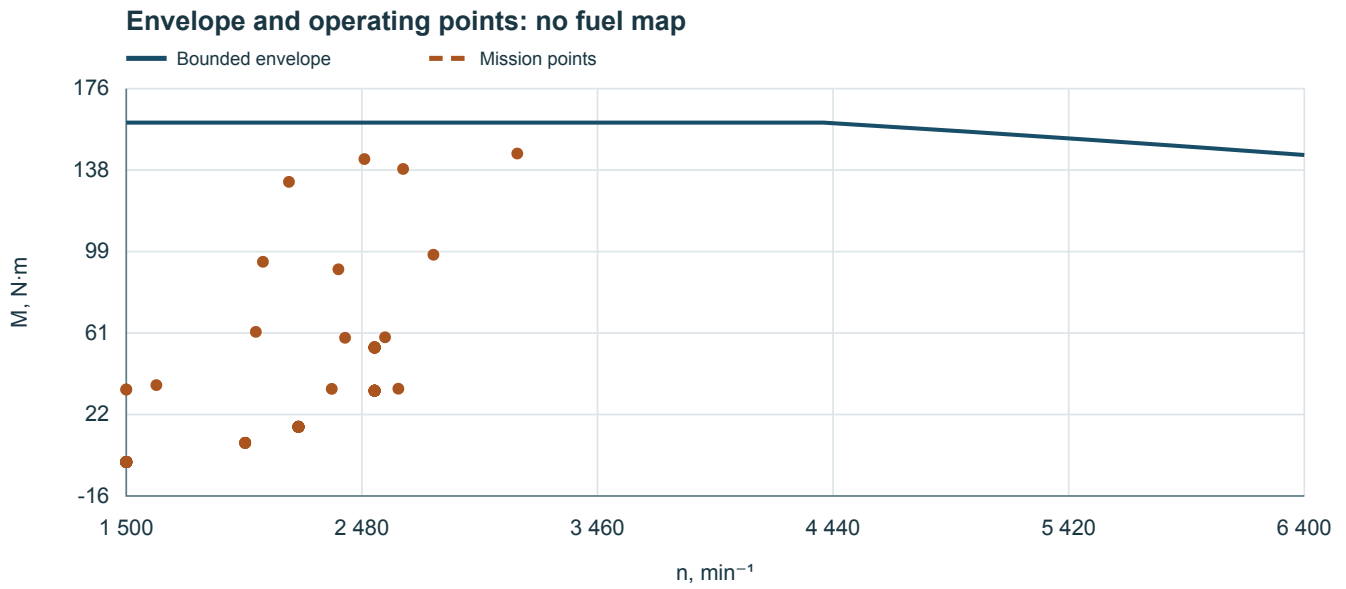
Symmetry of losses under torque reversal is assumed only for preliminary generating calculations. At Рмех < 0, Рел = Рмех + L; regenerative efficiency is therefore not calculated by dividing by motoring efficiency. An imported map is not mirrored automatically: generating operation requires negative-torque data.

The supplied Brogen CSV files contain contour points: Eff_… headers specify efficiency, Loss_… headers specify losses in watts, and cells specify power in kW. Torque coordinates are obtained from $M = 1000Р_{кВт}/(\pi n/30)$, only for n > 0. Empty and zero boundary points were not used as loss measurements. A trial fit produced unacceptable contour consistency; the dataset is retained with the conclusion rejected-for-default-use. This does not prove a manufacturer error: digitization, domain and map conditions need checking [6].

> No external numerical maps have been imported. Only explicitly described built-in models are active.



Motor map: only the coloured region is supported by digitised contours. Grey areas and generating operation are assumptions; this is not an EPA map.



Envelope and mission points. 1ZR has no fuel map; GG2W is a power-based approximation, not a measured 2D map.

4. Component selection and limits

The catalogue distinguishes EMRAX 208, 228, 268-C and 348, using only medium-voltage (MV), liquid-cooled (LC) versions. For the 228 MV/LC: 124 kW short-time S2 30 s, 68 kW continuous S1, 220/118 N·m, and 630 V for peak power. The 75 kW figure on the cover is not assigned to the LC version: it refers to different cooling. Cover and table values for the 348 conflict; the code retains the page 2 table and a warning [4,5]. The supplied v1.7 datasheets warn of a transition to a new generation; this example does not establish current availability for purchase.

Torque is constrained simultaneously by Tпаспорт, Kt·Iфаза,RMS and Pпаспорт/ω. The constant-torque/constant-power envelope is an approximate upper bound. Voltage limits, field weakening, actual inverter capability and temperature may reduce available torque. Battery DC current is determined separately and is not equated with RMS phase current.

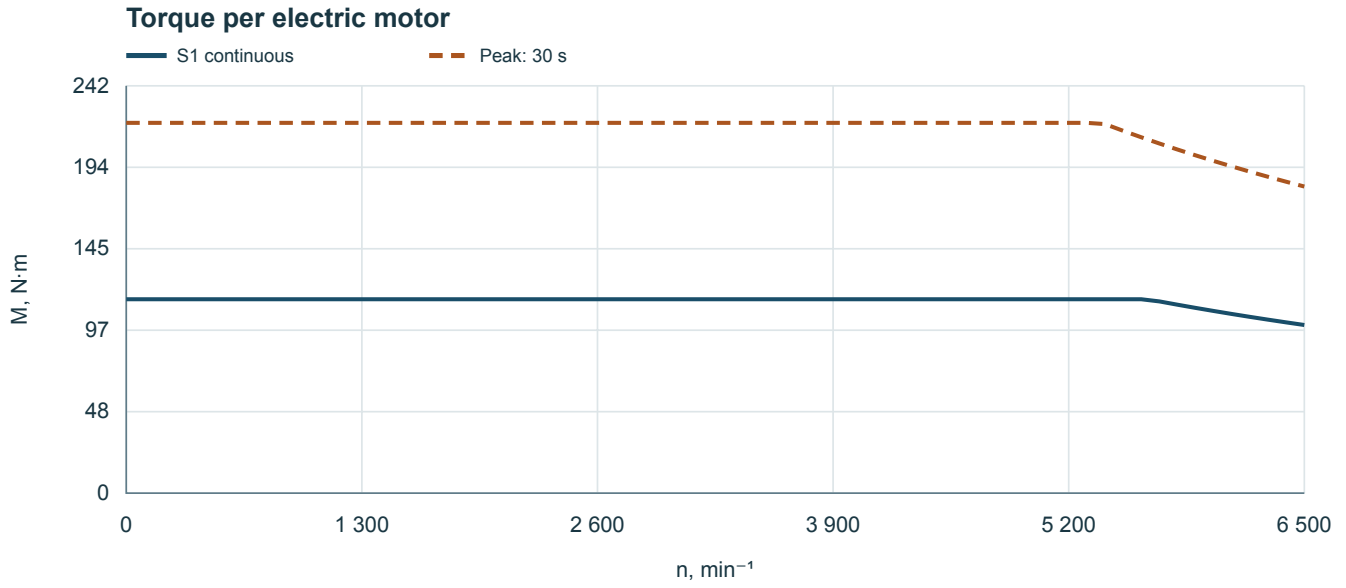
No numerically identified map is available in this release for the 208, 268-C or 348. An explicitly labeled assumption of η = 0,92 is used with a copper-loss floor. EMRAX 228 coefficients are not scaled to other motors. A rated “up to 96%” is not constant efficiency at every operating point.

Continuous limits are used in the mission. The peak characteristic is only a short-time S2 30 s envelope: it is not accumulated into unlimited operating time and does not imply that a thermal calculation has been completed. Warm-up, cooling, repeated peaks and derating are not modeled.

BEV and series layouts use one prescribed transmission ratio. P2/combustion-engine layouts use an assumed five-speed gearbox with ratios 3,5; 2,1; 1,4; 1,0; 0,8 and a separate final drive. Gear selection considers available force and proximity to the assumed operating speed. This is not a production shift map. Inertial gear-change transients are omitted. Multiple identical electric machines are represented at a common speed and transmission ratio; independent wheel-torque distribution is not calculated.

In P2, the combustion engine disconnects below its stable operating speed and the electric machine provides launch. The comparison combustion-engine layout assumes idealized low-speed clutch slip with separate accounting for heat losses. Without the required data, clutch life and exact acceleration time are not calculated. During braking, the engine in the comparison layout is disconnected from the wheels; with positive auxiliary demand, it runs at minimum speed to supply that demand. Fuel cut-off and discharge of the low-voltage battery are not modeled in this simplification.

Electric machine	S1 / S2, kW	S1 / S2, N·m	VDC for peak	nmax	Map in project
EMRAX 208 MV / LC	47 / 86	75 / 150	420	7000	η = 0,92, assumption
EMRAX 228 MV / LC	68 / 124	118 / 220	630	6500	Fit of two contours
EMRAX 268-C MV / LC	91 / 210	205 / 500	830	4500	η = 0,92, assumption
EMRAX 348 MV / LC	120 / 174	360 / 1000	830	4500	η = 0,92, assumption
EMRAX 228 HV CC — 2020 contour bands	62 / 109	96 / 230	680	5500	Numerical map: motor-only



Envelopes of one MV/LC motor with torque, power and phase-current limits.

5. Traction and energy calculation

The required force is F = mδa + mgfcosβ + ρCdA(v + vвітру)|v + vвітру|/2 + mgsinβ, where β = arctan(grade/100). Wheel power is Pк = Fv. All internal quantities use SI units; the interface speed in km/h is converted by dividing by 3,6, and rotational speed in min⁻¹ by multiplying by π/30. Positive wind speed represents a headwind [3,9].

For traction, Pвал = Pк/ηтр; for braking, electric-machine shaft power is Pвал = Pк·ηтр. Gearbox losses are nonnegative in both directions. Available traction force is limited by μ·zвед·mgcosβ. Total braking is limited by μ·mgcosβ; regenerative braking is additionally limited by driven-axle loading. This is a simple estimate, not an ABS/ESP or load-transfer model.

Parallel layout: Pк = ηтр(PДВЗ + Pм), only for positive total power flow. Series layout: Pген,DC = ηгенPДВЗ, Pбат = Pм,DC + Pдоп − Pген,DC. The battery may assist the generator or accept its surplus within the permitted limits. When the upper SOC limit restricts charging, generator power is reduced; surplus energy does not disappear from the balance.

Actual battery flow satisfies R_{int}: Uк = Uoc − IR, Pбат = UкI, I = 2Pбат/[Uoc + sqrt(Uoc² − 4RPбат)]. Positive I means discharge. Current, terminal voltage, discriminant, SOC and the charge available during a step are constrained. SOCнов = SOCстар − IΔt/(3600CA·h). For R = 0, the same stable expression is used without division by R [3].

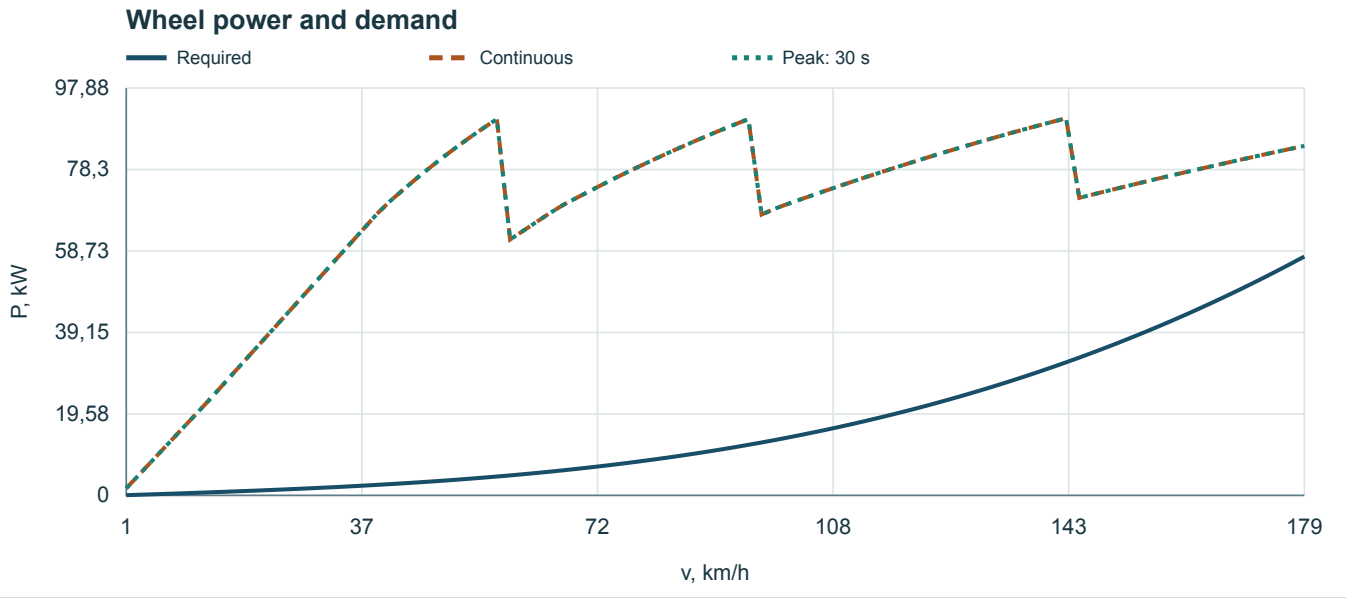
Constant Voc and R, with CA·год = 1000ЕкВт·год/Voc, define an assumed equivalent battery. I²R losses and the change in chemical energy UocI are accounted for separately from terminal energy. There is no Voc(SOC,T) map, dynamic polarization, aging, individual-cell voltage model or thermal validation. A displayed voltage below the motor's rated reference is explicitly flagged rather than hidden.

Braking demand is divided between permitted regeneration and friction braking. Energy charged from the engine in P2 is not counted as regeneration from vehicle motion. At high SOC or zero permitted charging current, the friction-braking share increases [9].

The hybrid controller is illustrative: SOC hysteresis of ±0,02 around the target and a wheel-power threshold. Fuel and electricity are shown separately. Lower petrol consumption while discharging the battery does not imply higher overall efficiency. A valid HEV comparison requires equal final battery energy or a documented SOC correction; no automatic “equivalent fuel” correction is invented here.

> F = 432,52 N
> Pк = F·v = 432,52·25 = **10,813 kW**
> Pвал,потр = Pк / ηтр = 11,382 kW

Condition / result	Value
Speed / gradient / acceleration	90 km/h; 0%; 0 m/s²
Available at the selected point	10,813 kW at the wheels
Deficit	0 kW
Battery DC current	0 A
Steady-speed BEV estimate, $SOC_0 \rightarrow SOC_{min}$ reserve	Not calculated for this layout/operating point



Wheel demand and available power. Initial SOC; peak is short-term, not continuous.

6. Scenario results and charts

The initial example is an author-defined 600 s cycle with 18 reference points, not WLTP, FTP75 or an EPA certification cycle. Speed is interpolated linearly; power flows are integrated with an internal step no greater than 1 s. A custom CSV uses columns time_s,speed_kmh,grade_pct. Time must increase and speed must be

nonnegative.

First check mission completion, energy deficit, time outside the approximation domain and SOC change. Only then analyze specific consumption. If the mission is incomplete, the reported consumption sums the loads actually served at the prescribed speed; it is not a prediction of real driving range.

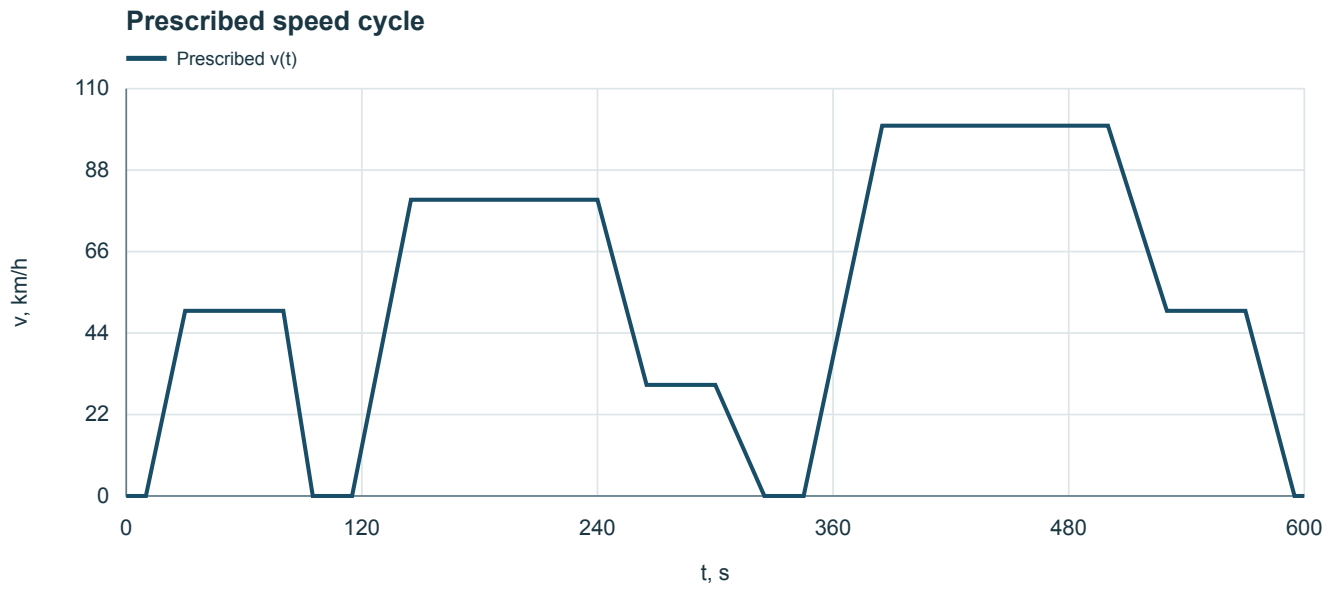
The power-selection chart shows wheel demand on the specified gradient and available continuous/peak power at initial SOC. These are instantaneous envelopes under current electrical and mechanical limits, not proof that peak power can be sustained along the entire route.

A range estimate is available only for BEV: constant speed on a level road, current auxiliary demand and available energy from initial SOC to the lower limit. Gradient and acceleration from the calculation point are not carried into this mode. The estimate does not account for battery temperature changes, variable speed or aging; it is not a rated driving range.

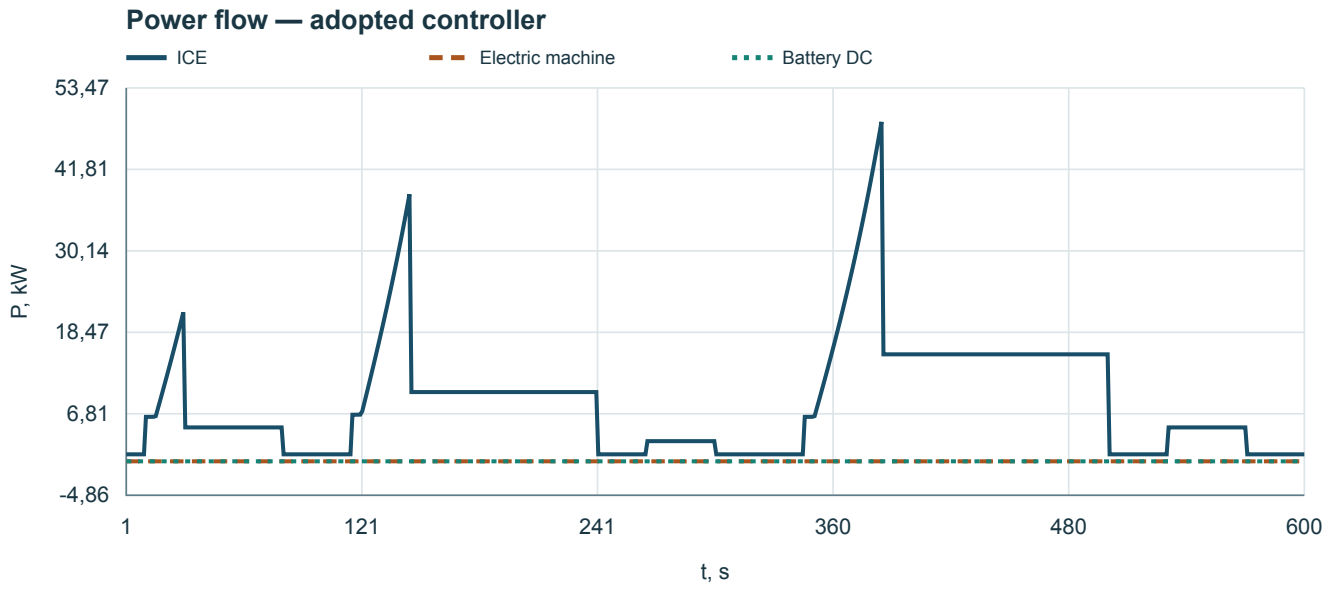
Calculated scenario

Mission supplied within the adopted limits. This status does not assess physical-model error.

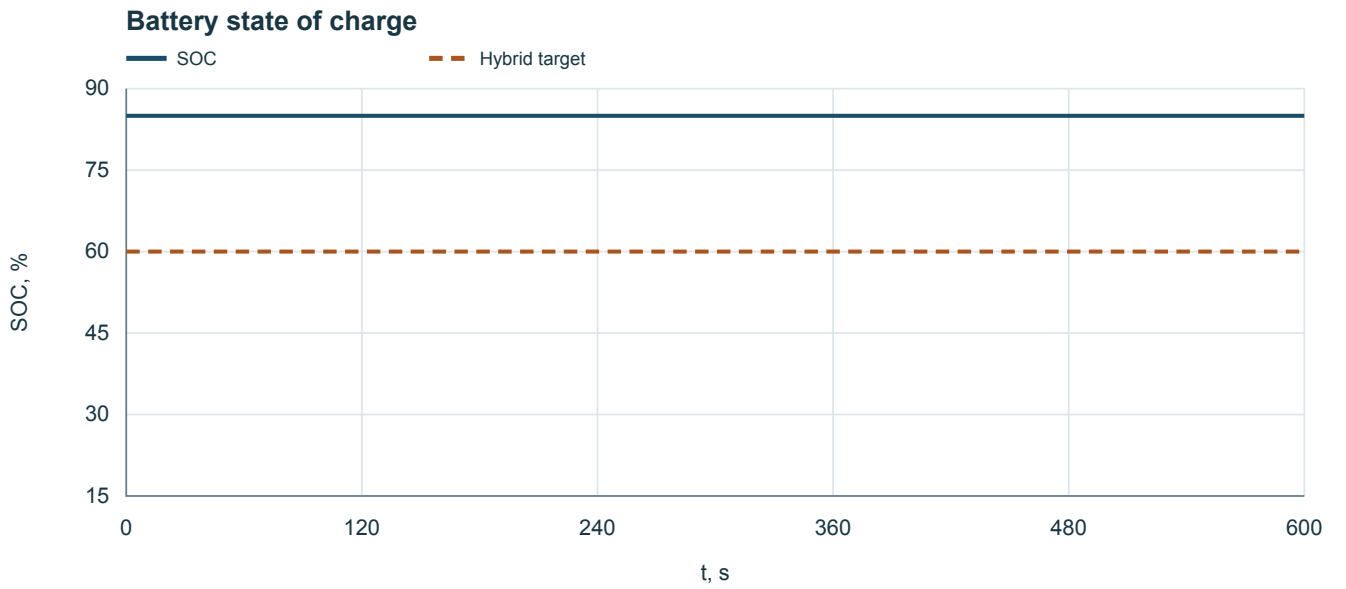
Quantity	Result
Speed schedule	Author-defined 600 s example
Layout	Combustion engine — comparison configuration
ICE	1ZR-FAE — two-point output-torque alignment
Motor	Not used
Mission distance / duration	9,264 km / 600 s
SOC initial → final	85 → 85%
Fuel	Fuel is unknown at some operating points; no complete consumption value is substituted
Net battery-terminal energy	0 kWh; 0 kWh/100 km
Battery chemical energy	0 kWh
Battery map / evidence	Constant Voc/R — adopted model
Battery temperature: start → end / max	Not calculated in the steady-state model
Battery I^2R losses	0 kWh
Recovered mechanical shaft energy	0 kWh, before electrical losses
Electrical regeneration at the DC bus	0 kWh
Accepted by the battery during braking, at terminals	0 kWh, after auxiliary demand
P2 charging from the engine through the electric machine	0 kWh, mechanical portion
Friction braking	0,22 kWh
Unserved demand / time	0 kWh / 0 s
Outside contour fit / generating assumptions	0 s from 600 s
Terminal voltage below the motor peak reference	0 s
HEV SOC numerical criterion	$\|\Delta SOC\| < 0{,}005$; this is a numerical criterion only



Prescribed mission speed. This is an author-defined example or imported cycle, not a verified achieved speed when a deficit exists.



Mission power flows; positive battery sign means discharge, negative means charging.



SOC change. Different final charge prevents comparing hybrids solely by litres of gasoline.

7. Checks, reproducibility and further data

Automated tests check rpm/rad·s⁻¹ conversion, map nodes and gaps, rejection of imported-map extrapolation, mechanical and DC balance, losses R_{int}, SOC limits, regeneration with a full battery, failure under insufficient power, preservation of unknown fuel consumption, and separation of training and verification groups.

Passing these tests confirms implementation of the stated mathematical model, not agreement with any production vehicle. Physical validation requires matched operating conditions and measurements of torque, speed, fuel flow, DC voltage/current, temperatures and final SOC.

Source files are organized by purpose: powertrain/manuscript contains small authored fragments; powertrain/src contains pure models and publication code; powertrain/data contains sourced data; powertrain/scripts reproduces fitting; tests contains checks. index.html, build/* and native JSON are generated; manually editing these derived files does not change the authored sources.

A saved powertrain-workspace/1 contains parameters, the cycle and explicitly imported maps. Results are recalculated when it is opened. Editing a field marks the result stale and blocks export until recalculation succeeds. Native engdoc exports text, scalars and explicit numerical snapshots; the Text_work editor does not independently run the vehicle integrator.

The next evidence-based step is to import an actual numerical EPA package for the selected engine/electric drive, verifying its system boundary, together with independently measured operating conditions and a cycle. Full 1ZR calibration requires the full-load curve, part-load conditions, fuel flow and indicator data; two rated maxima are not a substitute.

For further OBD calibration, each channel needs a unit, a PID/logger-calculation source, time and a validity status. Split data by trips, not adjacent rows. Do not treat MAF/lambda, Fuel Rate and Engine Power as independent channels until their definitions are established. The battery needs Udc, Idc, temperatures and traction-battery SOC specifically; regenerative charging must be distinguished from engine charging. Without these channels, results remain estimates/diagnostics rather than demonstrated parameter identification.

Numerical checks for the current scenario	Residual
Mechanical balance, max	0 W
DC balance, max	0 W

Numerical checks for the current scenario	Residual
Battery energy: Еном·ΔSOC − ∫UocI dt	0 kWh
Coulomb-counting residual	0 Ah
Thermal residual (SOC/T model only)	— kWh
Time with unknown fuel consumption	600 s
First uncovered point	{"time_s":0,"engineRpm":1500,"engineTorqueNm":0,"reason":"fuel-unknown"}
Map rejections	0
Unserved auxiliary demand	0 kWh

8. Requirements, forward acceleration and selection advisor

Forward acceleration integrates m·δ·dv/dt = Fдоступна − Fкочення − Fповітря − Fухилу. Traction is constrained by torque, power, speed, phase/DC current, battery and grip. At zero speed, torque is possible without mechanical power; copper losses remain.

Peak operation is allowed only for the shorter of the user-entered duration and the rated S2 duration. S1 is used afterward. This is a bounded single acceleration event, not a thermal model of repeated accelerations. Gear shifts are instantaneous; load transfer and detailed tire dynamics are not modeled.

Checks cover acceleration, achievable steady maximum speed at initial SOC, the specified gradient over the specified duration and, when required, electric range at constant speed. The latter is a preliminary estimate with constant Voc/R, not a WLTP result. Mission completion is checked separately.

The advisor searches a finite grid of up to 48 motor–transmission–battery configurations for the selected architecture and unchanged combustion engine. Configurations meeting both requirements and mission conditions rank first, followed by smaller assumed battery capacity and fewer motors. This is a transparent compactness rule, not cost optimization or production-component selection.

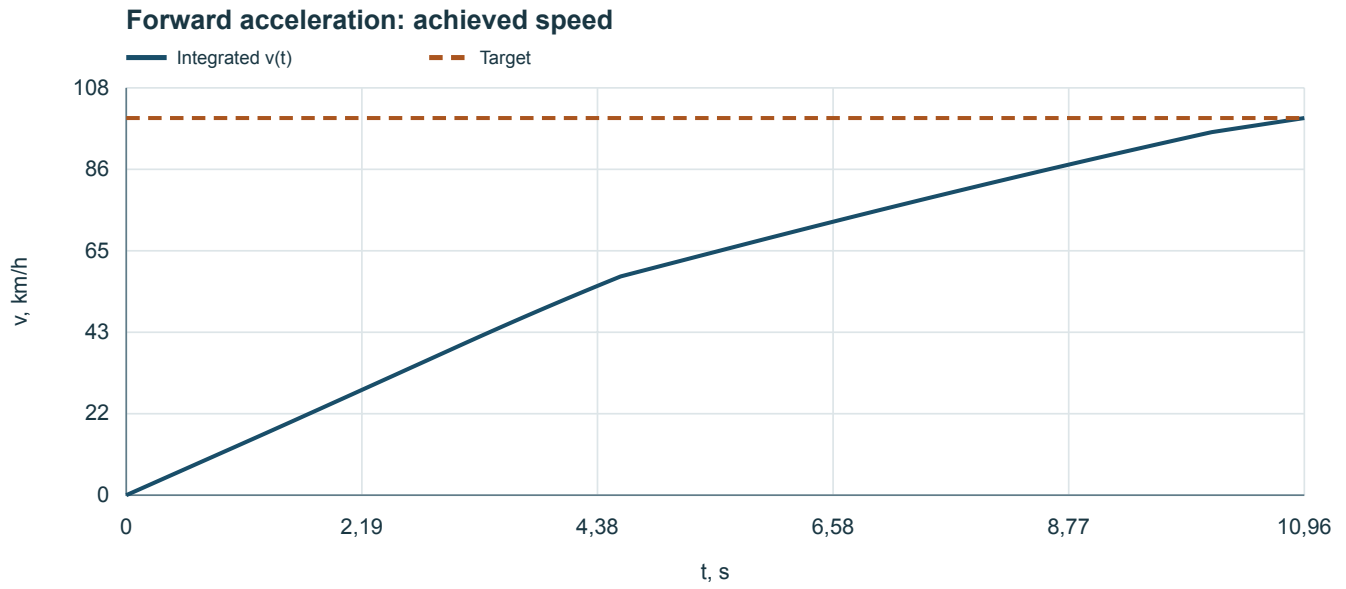
Assumptions for R_{int}-battery and inverter boundaries are explicitly identified. Operating points outside the map and generating assumptions remain warnings. Passing model selection does not establish experimental suitability.

Requirements met within the adopted model

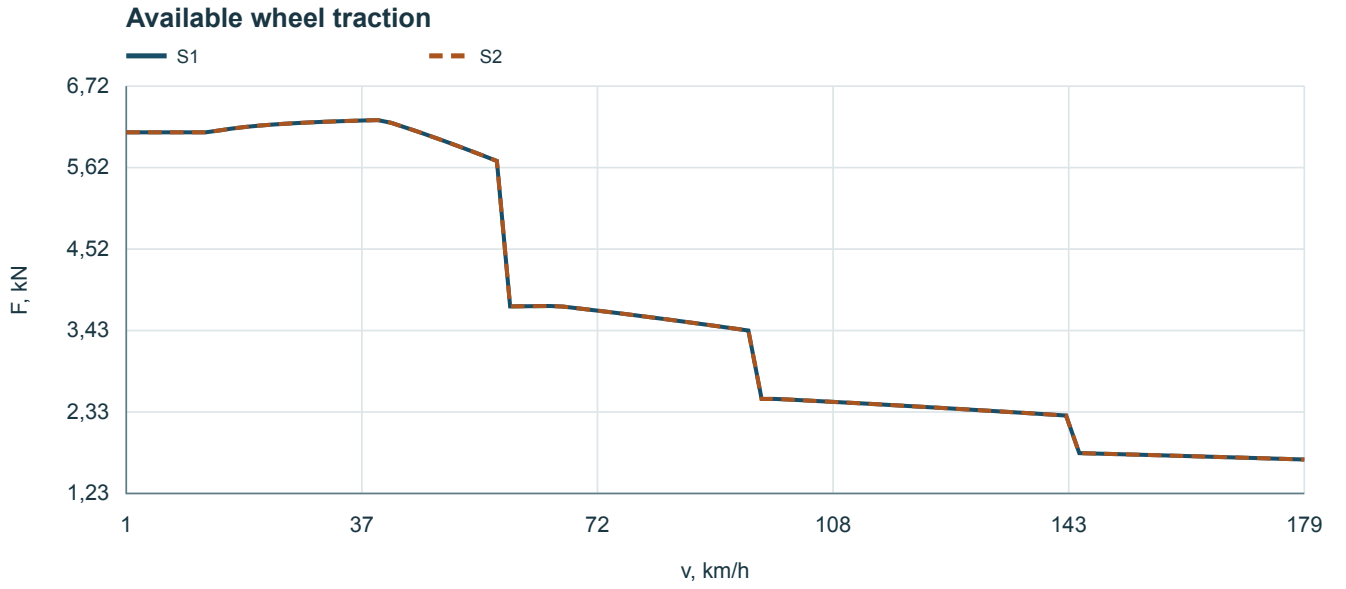
Check	Calculation	The available adhesion cannot satisfy	Status
Loaded voltage is no lower than the tabulated reference	1 yes=1	1	Satisfied
Constant 130 km/h at initial SOC	25,2 kW at the wheels	25,2	Satisfied
Acceleration 0–100 km/h	10,96 s	15	Satisfied
Gradient 10% / 40 km/h / 300 s	0 s deficit	0	Satisfied
Electric range at 90 km/h	—	0	Unspecified

Forward acceleration: checking	Value
Achieved speed	100 km/h
Acceleration distance	179,57 m
Final SOC	85 %
Integration step	0.1 s
Kinetic energy − integral of surplus force	0,00000001 J
SOC after separate gradient test	85 %

> Fuel consumption is unknown: this engine has no fuel map.



Achieved speed in forward acceleration, not a prescribed cycle. Peak duration is limited, followed by S1; shifts are instantaneous.



Available wheel traction at initial SOC and the selected limits. S2 is not a continuous characteristic.

9. Numerical maps and independent-trip checks

A further attempt was made to obtain EPA numerical ZIP packages; file transfer failed in this environment. A methodology link is not labeled as an imported map. The package contains controlled importers for normalized CSV/JSON and a failed-download log; synthetic tests are not described as EPA data.

An imported map must identify the component, units, component boundary, source and reference conditions. Fuel maps are not transferred between engine codes; loss maps are not transferred between motors. A missing cell is not zero fuel consumption. Interpolation outside the rectangular domain or across a gap is rejected.

For the independent OBD experiment, a separate time-series check is prepared with trip splits, units and signal provenance. Raw trips are absent here; no new GG2W training or independent validation was performed. Format examples used in tests are synthetic and are not mixed with the user's measurements.

Calibration of the 720° cycle, Voc(SOC,T), R(SOC,T), thermal limits and production hybrid controller remains separate work requiring suitable data. Existing OBD coefficients are retained unchanged.

Current parameter register

Rated motor parameters are separate from adopted vehicle, battery and controller conditions. Changing a baseline value is a user choice, not new manufacturer evidence.

Parameter	Unit	Value	Status
Powertrain architecture		Combustion engine — comparison variant	Adopted
Combustion-engine model		1ZR-FAE — two-point torque alignment (no fuel map)	Adopted
Engine brake-power limit	kW	97	Adopted
Engine torque limit	N·m	160	Adopted
Lower engine operating speed	min^{-1}	1 500	Adopted
Upper engine operating speed	min^{-1}	6 400	Adopted
Electric motor, MV/LC only		EMRAX 228 MV/LC — contour approximation	Adopted
Number of identical motors		1	Adopted
Separate inverter efficiency		0,97	Adopted
Peak phase current per inverter	A RMS	400	Adopted
Continuous phase current per inverter	A RMS	200	Adopted
Maximum DC power per inverter	kW	150	Adopted
Calculation mass	kg	1 600	Adopted
Aerodynamic drag coefficient		0,29	Adopted
Frontal area	m^2	2,2	Adopted
Rolling-resistance coefficient		0,012	Adopted
Air density	kg/m^3	1,225	Adopted
Rolling radius	m	0,31	Adopted

Parameter	Unit	Value	Status
Headwind (negative means tailwind)	m/s	0	Adopted
Equivalent rotating-mass factor		1,05	Adopted
Adopted adhesion coefficient		0,8	Adopted
Fraction of normal load on driven wheels		0,6	Adopted
Permitted duration of one peak acceleration	s	30	Adopted
Required continuous maximum speed	km/h	130	Adopted
Final acceleration speed from 0	km/h	100	Adopted
Acceleration-time limit	s	15	Adopted
Required sustained gradient	%	10	Adopted
Sustained-gradient speed	km/h	40	Adopted
Gradient-driving duration	s	300	Adopted
Required electric range (0 — do not check)	km	0	Adopted
BEV / series-layout ratio		5,5	Adopted
Final drive P2 / ICE		3,7	Adopted
Mechanical-transmission efficiency		0,95	Adopted
Auxiliary demand	kW	1	Adopted
Nominal battery energy	kWh	48	Adopted
Constant open-circuit voltage Voc	V	700	Adopted
Minimum terminal voltage	V	630	Adopted
Maximum terminal voltage	V	800	Adopted
Pack internal resistance	Ω	0,12	Adopted
Maximum DC discharge current	A	250	Adopted
Maximum DC charging current	A	100	Adopted
Initial SOC		0,85	Adopted
Lower limit SOC		0,15	Adopted
Upper limit SOC		0,9	Adopted
Hybrid-controller target SOC		0,6	Adopted
Engine-start threshold based on wheel-power demand	kW	25	Adopted
Preferred engine operating point in the series layout	kW	45	Adopted
Generator electrical-power limit	kW	60	Adopted
Generator efficiency including rectifier		0,93	Adopted
Charging power margin	kW	4	Adopted
Calculated speed	km/h	90	Adopted
Gradient at the calculation point	%	0	Adopted

Parameter	Unit	Value	Status
Acceleration at the calculation point	m/s²	0	Adopted

Sources and use status

[1] Previous ICE 0.2.0 and Toyota register in data/sources.json. 1ZR-FAE: rated performance points only. The original 720° module is retained without new physical calibration.

[2] The user's experimental project with an OBD-II/MX+ logger. vehickle_math_models / vehicle_model_library_examples_v0.11.zip: vehicle_instances/mitsubishi-outlander-phev-2014-gg2w/calibrations/willans_engine_fuel_map_v1.json; config/outlander_2014.yaml. Coefficients and reported errors were carried over. MX+ calculates Engine Power; the Willans reference is a MAF/lambda estimate at AFR 14,1. Raw training trips are not supplied in this archive; independent test-bench accuracy is not claimed. Selected copies and SHA-256: reference/model-library and powertrain/data/model-library-provenance.json.

[3] The same library: willans-affine-fuel-model, battery-rint, battery-soc-coulomb-counting, motor-efficiency-map, regenerative-braking-limit, longitudinal-force-balance, EPA base/transient catalogues. These models were checked, not every archive file. The JS implementation is separate; the library simulator is not represented as a passed test.

[4] EMRAX 228, supplied datasheet v1.7: p. 1 — digitised contours; p. 2 — MV/LC, Rs and Kt. Chart conditions and the tabulated version are not declared identical. Copy: reference/electric; points/coefficients: powertrain/data/emrax228-loss-fit.json.

[5] EMRAX 208, 268-C, 348, supplied datasheets v1.7, tables on p. 2. For 348: promotional values on p. 1 conflict with the table; the MV/LC table is adopted with an explicit warning. Other sizes use assumed efficiency only in this implementation.

[6] Brogen, Centralized Electric Axle for Trucks, Dec 2025, p. 5, and two supplied contour CSV files. Raw CSVs are preserved. The dataset is excluded from default consumption prediction after failing its consistency check. The Brogen e-motor catalog June 2026 is registered, but its promotional maxima do not replace a map for the selected SKU.

[7] EPA — Combining Data into Complete Engine ALPHA Maps, methodology page checked on 25.09.2026. Original ALPHA numerical packages are absent; the restricted literal adapter was checked against synthetic fixtures, not original ZIP files. After a separate import, active sources appear in section 3.

[8] EPA — Combining Data into Complete eMotor ALPHA Maps, checked on 25.09.2026. EMOT / EDU / BISG system boundaries differ; our restricted adapter does not import every ALPHA package.

[9] Ehsani M., Gao Y., Longo S., Ebrahimi K. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles. 3rd ed., 2018. Series/parallel operating modes and power allocation were used, particularly chapter 9 and regenerative-braking limits. Numerical book examples were not adopted as specifications of the selected assemblies.

[10] EPA Dynamometer Drive Schedules, checked on 25.09.2026: Data/EPA contains the user's UDDS/HWFET/US06 copies with verified checksums. These are speed schedules, not component maps or the complete regulatory procedure.

[11] MathWorks, Battery Equivalent Circuit: principles of OCV/R dependence on SOC/T. Additional author-defined R_{int}/T module: ohmic heating and linear cooling only, without ageing, polarisation or entropic heat. Numerical parameters are specified through a separate source.

Other supplied books by Enge/Enge/Zoepf, Zhang and Zheng, Volkswagen SSP 499 and Introduction to Internal Combustion Engines are registered for future expansion. Registration does not imply complete reading or numerical calibration against them. Book files are not duplicated in the software archive.